Jacqueline Pantoja

Baltimore City Traffic Violations April 2012

Upon first inspection of the data, I noticed that the variables with the least amount of missing data were the date of the violation, make of the vehicle model, state where the violation took place, description of the violation, the fine amount, and the individual's outstanding balance; the analysis was conducted with this data.

To get an idea of what to trends to expect, I filtered the data for a single month in 2014, and compared fine amounts stratified by outstanding balance tertiles. Among each category, most fines were in the low $30 range, while low balance folks had the rest of the fines ranging from 100-300 dollars, while the high balance group ranged from 100-500 dollars. Those with a middle-ground outstanding balance had fines of $32—with two exceptions taking place at night: $52 overnight parking violations and $77 parking in a transit stop—for the month of April of 2014. This prompts the question, what is the most likely reason for these violations and what conditions make this more likely (e.g. expired tags, make and model, time of day, etc.)

When looking at the distribution of parking violations given by time of day (dichotomized as day=AM or night=PM), the counts differ by around 66 parking violations more for violations written up at night (out of 584 total), and we also see a substantial increase in Parking

TimeOfDay	Description	
day	All Other Parking Meter Violations	91
	All Other Stopping or Parking Violations	25
	Exceeding 48 Hours	6
	Expired Tags	75
	No Stopping/Standing Not Tow-Away Zone	55
	Passenger Loading Zone	7
night	All Other Parking Meter Violations	165
	All Other Stopping or Parking Violations	18
	Blocking Garage or Driveway	2
	Exceeding 48 Hours	6
	Expired Tags	75
	No Stopping/Standing Not Tow-Away Zone	57
	Passenger Loading Zone	2

Meter Violations (+74). We can conclude that one large contributor to the differences in Fines given are due to time of day and the increase in parking violations taking place at all hours of the afternoon compared to the limited hours of the morning where fines can be given (usually 8-12).

Additionally, we can also observe the impact of having expired tags as the reason for a traffic violation contributes much more during the day even when the total number of citations is exactly the same (75 for both times of the day), and a total of 62 traffic violation are filed for folks who did not have their tags expired.

Out of these sixty-two folks, those with Nissans, Fords, or Hondas had the highest counts of mistaken issued traffic violations at 5, 7, and 11 counts respectively.

The last observation made was looking at the overall burden of mistaken traffic violations for expired tags for folks who did not have expired tags. As expected, expired tags make up over 50% of the reason for the traffic violation fine among all available citations issued for folks with expired tags (56%, 23 thousand out of 42 thousand). For those with mistaken traffic violations for expired tags, about 5 percent of folks were fined for expired tags (4.74%, 7 thousand out of 146 thousand).

*Data used in this analysis was provided from this site, and can be replicated using the provided ipynb notebook.